

Introduction to Object-Oriented Programming

CS XXX: Elementary Programming

Bob Smith • January 01, 2026

ILLINOIS TECH

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Objects

Color Demos

- This is an example of red text.
- This is an example of green text.
- This is an example of blue text.
- This is an example of **bold** text

Logistic Regression

First Slide

A *slide* with a title and an **important** information.

1. Hello world
2. Goodbye!

```
class Person:
    def __init__(self, name: str, age: int):
        self.name = name
        self.age = age
```

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Federated Averaging

$$\omega^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \omega_k^{t+1}$$

Hello, this is a function: $f(x)$.

Verbose

Theorem 1 (Euclid's Theorem)

There are infinitely many prime numbers.

We define Theorem 1 using the explicit form:

```
#theorem(title: "...", supplement:  
"...") [body]
```

Simple

Theorem 2 (Simple) There are infinitely many prime numbers.

We define Theorem 2 using the explicit form:

```
#theorem[<name>] [body]
```

Other Theorem Types

Theorem 1 (Name) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore.

Definition 2 (Name) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore.

Corollary 2.1 (Name) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore.

Property 3 (Name) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore.

Example 4 (Name) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore.

Lemma 5 (Name) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore.

Assumption 6 (Name) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore.

Conjecture 7 (Name) Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore.

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore. This concludes the proof. $\square$

Important Notes

Note

Remember that mathematical proofs should be both rigorous and clear. Clarity without rigor is insufficient, and rigor without clarity is ineffective.

Caution

When dealing with infinite series, always verify convergence before discussing other properties.

Mathematics is the queen of sciences, and number theory is the queen of mathematics. — Gauss

1. Hello
2. Goodbye!

Edge Purpose and Definition

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An **title-free slide** can be created by the following:

```
#slide(title: none, ...)
```